DRILL 10

RURAL WATER SUPPLY

NFPA 1001 Level 1 (JPR 4.3.15) / FUS STSS Standard

RURAL DROP TANK DEPLOYMENT & TANKER DUMP OPERATIONS

⏱ **Time:** 60–75 min | 🚒 **Apparatus:** Pumper & Tanker (Mobile Water Supply) | 🦺 **PPE:** Full Gear or Work Gloves/Boots

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Drop-Site Crew & Tanker Driver

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to remove a 1,500 to 3,000-gallon portable folding drop tank (Porta-Tank) from apparatus racks, deploy it on level ground, stage a low-level strainer, set up a jet siphon between dual tanks, and guide a tanker into position to dump 1,500+ gallons in under 90 seconds. Mastering this drill is mandatory to pass and maintain **Fire Underwriters Survey (FUS) Superior Tanker Shuttle Service (STSS)** accreditation in New Brunswick.

📦 TOOLS & APPARATUS SETUP

- Pumper with hard suction hose (6-inch / 150mm) and low-level strainer with air-bleed float.
- Tanker with portable folding drop tank (2,000–3,000 gal) and 10-inch rear or swivel side dump valve.
- Jet siphon attachment and 1.75" discharge line.
- Ground tarp to protect drop tank liner from sharp gravel/debris.

⚠ SAFETY WATCH-OUTS & STOP RULE

⚠ **Backing Apparatus Hazard:** The tanker must have a designated backing guide wearing a high-visibility vest. The guide must stay in the driver's left-side mirror at all times!

⚠ **Pinch Points:** Keep hands clear of folding tank scissor frames when opening. Frame hinges can pinch or crush fingers.

⚠ **Water Weight:** A 2,500-gallon tank weighs over 25,000 lbs (11,300 kg) when full! Once filled, it cannot be pushed or moved without draining.

🛑 **THE STOP RULE: Shout "STOP!" immediately if the backing tanker loses sight of the guide or if the tank begins to collapse due to uneven ground slope.**

PHASE 1: SITE SELECTION & TANK DEPLOYMENT (15 MINUTES) — SLOW DRY RUN

1. **Site Selection:**
   - Choose flat, firm ground adjacent to the attack pumper.
   - Ensure adequate lane clearance (minimum 12 feet) for incoming tankers to pull past, dump, and exit without turning around.
2. **Deploying the Folding Tank:**
   - Firefighters lay a heavy protective canvas/vinyl ground tarp on the road surface to prevent gravel punctures.
   - Release retaining straps and lift the folding tank off the tanker rack (2–4 firefighters).
   - Stand the folded tank vertically on its bottom rails.
   - Two firefighters grasp opposite corners of the frame, pull outward smoothly until frame locks open in a square/rectangle.
   - Push the internal vinyl/hypalon liner down into the corners; ensure the drain sleeve is cinched tight and tied to the top rail!

DRILL 10: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Rural Water Supply • Practical Application

PHASE 2: LOW-LEVEL STRAINER & HARD SUCTION STAGING (15 MINUTES) — BUDDY COACHING

1. Connect low-level strainer to 10-foot section of 6-inch hard suction hose.
2. Lower strainer into the drop tank until it sits flat on the tank floor.
3. Secure the suction hose to the tank frame using a strap or short rope (prevents the strainer from flipping upside down when the pump pulls a vacuum).
4. Connect hard suction to the pumper's master intake.

PHASE 3: RAPID TANKER DUMP & JET SIPHON (25 MINUTES) — TIMED / STRESS

1. **Guiding the Tanker to the Dump Site:**
   - Spotter directs the tanker driver using standard hand signals.
   - Position the tanker's 10-inch dump chute directly over the lip of the drop tank.
   - Tanker driver sets air brake and trips the air-actuated or manual dump valve.
   - Water dumps in a high-volume torrent (aim for 1,500–2,000 Imperial gallons offloaded in 60–90 seconds).
2. **Dual-Tank Jet Siphon Setup (For Continuous Flow):**
   - When setting up a second drop tank for high-flow shuttle operations, place Tank 2 end-to-end with Tank 1.
   - Place a **Jet Siphon** tube between Tank 2 (dump tank) and Tank 1 (drafting tank).
   - Connect a 1.75" hose line from the pumper discharge to the jet siphon inlet.
   - Flowing 50–75 PSI into the jet siphon creates a strong Venturi vacuum, transferring water continuously from Tank 2 into Tank 1 at over 500 GPM!

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Drain sleeve untied:* Dumping 2,000 gallons into a tank with the drain sleeve untied means half your water runs down the ditch!
- ❌ *Setting tank on steep road camber:* If the road crown is too steep, water will spill over the downhill rail before the tank is even 60% full.
- ❌ *Strainer vortexing:* At low water levels (under 6 inches), swirling air vortexes form. Keep a floating ball, plastic lid, or rubber mat over the strainer inlet to break up whirlpools.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"How many seconds did it take from tanker stopping to dump valve open?", "Did the jet siphon keep Tank 1 level with Tank 2?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Drain sleeve opened to release remaining water.
- ☐ Liner wiped dry and swept clear of gravel/mud before folding.
- ☐ Frame folded neatly and secured back in the apparatus mounting bracket.
- ☐ Suction hose and strainer drained and mounted.
- ☐ Tanker refilled 100% from station fill line.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS